FUZE Software Quality Assurance Plan

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Revision History

Revision	Date	Prepared for	Comments
1.0	6 Oct 2026	Team FUZE	Initial Project Plan.
1.1	7 Oct 2026	Team FUZE	Added critical path, prototype status and remaining-work estimates (Section 4.6). Compared code size with the estimate. Risk ratings now kept in the Risk Management Plan. Corrected rounding. GitLab replaces GitHub.

Document basis

ThesectionorderfollowstheSC3040sampleprojectplan.FUZEcontentisdrawnfromthefour documents below. Source references use the document key together with the original section, table, figure, use-case or specification identifier.

Key	Controlling source for this compilation	Main use in this plan
P	Project_Proposal.docx — Proposal for the Design of FUZE	Objectives, scope, six phases, M1–M6, team roles, D1–D12, resource estimates and sponsor communication.
U	UseCase.docx — Use Cases for FUZE, Version 2.0	UC-1–UC-16, user workflows, exceptions and recorded open questions.
Q	Quality Plan.pdf — Software Quality Assurance Plan, Revision 3.0	Quality controls, DS-1–DS-16 evidence, reviews, testing, risks, configuration and records. Approval remains pending in this source.
S	System Requirement Specification.pdf — SRS, Revision 1.0	Functional and process requirements, rule semantics, interfaces, deployment requirements and OI-1–OI-22.

Planning status and limits

Dates are planned dates. Progress against them is recorded in Section 4.6, which gives the state of the prototype on 7 October 2026. Work-package identifiers group documented work and add no scope.

The 17 September meeting minutes were read for context only. Their short-term dates do not replace the four-document schedule. One assignment from them is adopted by team decision: Mohika builds component data and caching, with Armaan reviewing the adapters (proposal Table 7 gives this work to Armaan). The separate frontend schedule is not used. Source conflicts and missing inputs are retained in Sections 5 and 10 rather than silently resolved.

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1 Introduction

1.1 Project Overview

FUZE is a self-servicewebapplicationforwhole-boardconstraintvalidationofelectronicbillsof materials (BOMs). An engineer describes a circuit board in English, and the system derives component slots and power rails, sources real components and checks their relationships as a graph. The interface streams the run and displays the field, value and source behind each verdict. The project runs for fourteen weeks, from 20 August to 25 November 2026, with a seven-member team.

[P: Objectives; Project Management, Tables 5–7. S: §§3.2, 5.8, 7.3 and 7.8.]

The technical boundary is deliberate: a deterministic constraint engine determines compatibility; a language model interprets the brief, normalises catalogue text and selects a repair strategy from engine-computed legal options. The engine checks that choice before applying it and revalidates the board afterwards. It has no network client, model client or access to the system clock. The model does not issue compatibility pass/fail verdicts.

[P: Objectives and Technical Approach. S: §§3.2, 8.3, 8.9 and 11.7.]

The proposal sets four objectives: a deterministic and evidence-bearing rule engine; a sourcing pipeline that preserves each value's origin; a bounded, model-fenced repair loop; and a streaming application through which an engineer can inspect results both within a run and across completed runs. The project serves hardware engineers, junior engineers and student design teams. The proposal and SRS report fifteen preliminary interviews as the basis for refining these needs; the underlying interview records are not supplied with this plan.

[P: Previous Related Work, Objectives and Customer Needs. Q: §1.2. S: §4.1.]

1.2 Project Description and Scope

The application scope is organised around the sixteen documented use cases. Section 4.6 records which of these functions work in the current prototype.

Functional group	Included capability	Use cases
Account and session management	Registration, sign-in, session restoration and sign-out; guided introductory walkthrough.	UC-1, UC-5
Project management	Create, browse, rename and delete owned projects; browse the runs recorded under a project.	UC-2–UC-4
Board design and validation	Submit a brief; watch the trace and graph; automatic repair; answer an escalation; recover an abandoned/errored run; inspect a saved board without resuming it.	UC-6–UC-11
BOM and reference tools	Validate CSV or pasted MPNs; export a run's BOM; attach a manufacturer datasheet to supply a missing thermal value.	UC-12–UC-14

Functional group	Included capability	Use cases
Memory and analytics	Browse retained parts, boards and findings; review rule outcomes, repair effectiveness, recurring conflicts and supply exposure across completed runs.	UC-15–UC-16

[U: UC-1–UC-16. S: §4.2.4, Table 1; §§7.1–7.12. SRS OI-19 and OI-20 retain sponsor scope-confirmation actions for memory, walkthrough and standalone BOM validation; see Section 10.]

Each rule returns pass, fail, warning or unchecked as applicable. Missing information is named and is never replaced by a guessed electrical, thermal, dimensional or stock value. An unchecked result is not a pass, and a board containing unchecked results must not be described as fully validated. A conflict view includes every rule that ran on the affected part, including passing rules.

[S: §§7.3–7.4, 8.4 and 9.1–9.2. Q: §7.]

Constraint-engine scope

Rule	Required behaviour
R1 voltage_overlap	Check part.vmin ≤ rail.v ≤ part.vmax. Missing limits or an unassigned rail yield unchecked.
R2 current_budget	Fail when the sum of peak rail currents × 1.15 exceeds the source's maximum current. Missing current data are not treated as zero.
R3 thermal_dissipation	Evaluate regulator dissipation and temperature rise using the sourced thermal resistance. Identify any stated conservative continuous-operation assumption; missing required inputs remain explicit.
R4 interface_role_match	Check protocol family, complementary roles and compatible signal levels, or the presence of a level translator.
R5 pin_budget	Compare required peripheral pins with available controller pins; count a shared bus once. Missing pin counts yield unchecked.
R6 availability	Fail for insufficient stock; warn for NRND/obsolete lifecycle status. Freeze the sourced commercial snapshot within the run for repeatable replay.
R7 temperature_rating	Require the component's range to cover the requested operating range. An absent component range or requested range yields unchecked.
R8 footprint	Provide a warning, never a failure, for a package unsuited to the intended assembly process.
R9 energy_budget	Check battery capacity divided by continuous draw against required runtime when the necessary battery/runtime information is supplied.
R10 rail_coverage	Report a component without an assigned rail as unchecked, never as a pass.

[P: Table 2. S: §§7.4.1–7.4.10. Detailed arithmetic and input definitions remain in the SRS and D3.]

Two additional SRS advisory thresholds flag operation above 80% of a rated limit and a calculated temperature rise above 60°C. They are not additional registered rules and do not produce pass/fail results. The repair vocabulary consists of swap, change_topology, add_part, change_rail, relax_requirement and

escalate. A slot explicitly named by the engineer is restricted to swap, change_rail and escalate. At most three repairs are applied per slot; the SRS also caps add_part at three added slots per run. The policy governing relax_requirement remains an open issue.

[S: §§7.4.10, 8.7 and 8.9; OI-21. U: UC-8.]

FUZE validates at block level. It does not generate schematics, place or route a board, assign pins, generate netlists, simulate circuits or produce manufacturing files. Horizontal scaling is outside scope; the application uses a single worker for the event stream. Analytics is limited to the four descriptive views above: it does not forecast obsolescence or lifecycle transitions, create a composite health grade, independently rate a part's reliability, or compare different engineers' designs.

A fabricated test board and bench instruments are specified for validation measurements. They are test resources, not a hardware product to be delivered and not an extension of the application into board synthesis. Their resource and measurement details are not supplied; Section 5.3 records this limitation.

[P: Scope exclusions in Technical Approach; Table 3; Figure 4. Q: §1.2. S: §§5.5, 7.12 and 10.4.]

2 Project Organization

2.1 Team Structure

TeamFUZEhassevennamedroles.Mohikaownsprojectcoordinationandsponsorcontact.Sparsh is the technical design authority. Armaan and Lai Yi lead the backend and frontend work respectively. Samarth and Faye share quality management, with verification and validation separated. Jerral owns the release and infrastructure workstream. QA reports quality status and unresolved risks to Mohika independently of the author whose work is being assessed.

[P: Table 7. Q: §§3.1–3.3.]

2.2 Roles and Responsibilities

The responsibilities below retain the role allocation in the proposal and the quality responsibilities added in Revision 3.0 of the Quality Plan. Package leads in Section 4 coordinate the associated work; they are not assigned all supporting effort by default.

Member / role	Responsibilities and accountability
Mohika Project Manager	Schedule and budget tracking; sponsor liaison; status reports; change control; risk register; meeting minutes and approvals; component data and caching (JLCPCB and Mouser adapters, normalisation and the physical-property cache). Accountable for M1, M6 and sponsor communication.
Sparsh Lead Developer	Technical direction; final design authority; event/interface contract; constraint rules and resolution policy; architectural review. Reviews engine or contract changes. Accountable for M2, M3, DS-1–DS-4, DS-7 and DS-10.
Armaan Backend Developer	FastAPI interface, LangGraph orchestration, sourcing integration and review of the distributor adapters, datasheet ingestion, persistence and authentication. Accountable for M4, DS-5, DS-6, DS-9 and DS-13; supports DS-16 analytics computation.
Lai Yi Frontend Developer	React application, streaming client, component graph, conflict drawer, BOM table, account/project screens, accessibility and analytics views. Accountable for M4, DS-11, DS-15, DS-16 and the client side of DS-6; supplies CSV interoperability and client recovery/fallback evidence.
Samarth QA Manager — Verification	Unit/property tests, rule arithmetic cross-checks, regression boards, coverage gates and adversarial/injection corpora. Verifies DS-1, DS-4, DS-8, DS-14 and DS-16; supports DS-2, DS-3, DS-7 and DS-10 evidence.
Faye QA Manager — Validation	End-to-end browser tests, acceptance scenarios, ten-participant usability sessions and sponsor evidence pack. Coordinates M5 and the DS-1–DS-16 acceptance evidence.
Jerral Release Engineer	CI/deployment, environments and secrets, schema changes, monitoring, fixture recording, release tags and runbook. Accountable for DS-12 and the infrastructure aspects of DS-13.

[P: Table 7. Q: §§3.1.2 and 3.3; Appendix A.]

Every change requires an engineering approval from someone other than its author. Sparsh reviews changes touching the engine or interface contract. When QA staff author a change, another qualified reviewer verifies it. Developers fix defects; the QA roles remain responsible for verification/validation evidence and retesting.

[P: Project Management, paragraph following Table 7. Q: §§3.1.2 and 8.]

2.3 Team Communication

Communication	Frequency / format	Responsibility and required action
Written sponsor status report	Every two weeks, Thursday; one-page progress, decisions, blockers and next steps.	Mohika to sponsor and team. Normally informational; blockers are flagged.
Sponsor call	Every two weeks, Thursday; 30 minutes on status, decisions and risk.	Mohika chairs; Sparsh handles technical questions. Sponsor decisions are requested where needed.
Milestone review	M1–M6; 45–60 minutes. Deliverable circulated 48 hours in advance.	Full team demonstrates exit criteria; Mohika records decisions. Written sponsor sign-off closes the phase.
QA / risk / defect review	Weekly with the team, and at milestone gates.	Samarth and Faye report quality, defects and risks to Mohika; correctness or milestone threats are escalated immediately.
Change request	As required; written reason, scope, schedule and budget impact, and displaced work.	Mohika coordinates sponsor approval. No scope change is actioned without written approval.
Technical query	As required; email or shared issue-tracker thread.	Proposal Table 10 requests a sponsor reply within three working days. State the assumption to be used if no reply arrives.
Escalation	As required; direct call, followed by a written summary within 24 hours.	Mohika contacts the sponsor and requests an immediate response.
Repository / project-board access	Continuous sponsor read access from Week 1.	Progress and artefacts are inspectable without a separate request.
Final handover	Week 14; demonstration, training and artefact transfer.	Full team; sponsor acceptance sign-off requested.

[P: Table 10. Q: §§6, 8 and 14. S: §§6.1–6.2.]

The sponsor cadence above follows proposal Table 10 and Quality Plan Section 8. A nearby proposal sentence refers to a weekly call, and the SRS describes three-working-day technical support responses differently from the proposal's request for a sponsor reply. These wording differences are retained as communication clarifications in Section 10. No additional internal meeting day, messaging platform or support commitment is assumed.

3 Process Definition

3.1 Lifecycle Model

FUZEfollowsanincrementallifecycle.Itisphasedandmilestone-gated,withdevelopment,testing and release preparation overlapping. Each increment widens a working vertical slice and is closed by automated tests.

The proposal defines six phases. A phase remains open until its exit criterion is demonstrated to the sponsor and written sign-off is recorded. Frontend, backend and engine work proceed in parallel against a common contract. Unit, integration and end-to-end testing are used throughout development rather than deferred entirely to the final phase.

Phase	Weeks	Work and exit condition
P0 Planning	1–2	Requirements elicitation, interviews, freedom-to-operate search, tooling/repository setup. Exit: requirements baseline at M1.
P1 Concept Development	2–4	Evaluate data sources against rule fields, select/benchmark the model, storyboard the user journey and write a failing executable acceptance scenario. Exit: source-selection memo accepted and scenario written.
P2 System-Level Design	3–5	Define event contract, provenance-bearing part type, board graph and API surface. Exit: interface contract frozen at M2.
P3 Detailed Design and Implementation	5–11	Build engine, planner, sourcing, repair/orchestration, API and frontend bottom-up. Exit: live-data end-to-end vertical slice at M4.
P4 Testing and Refinement	10–13	Unit, integration, adversarial, browser, usability and performance work. Exit: specification evidence at M5, subject to the DS-12 observation-window issue.
P5 Production and Release	12–14	Deployment hardening, documentation, runbook, final report and demonstration. Exit: final delivery at M6.

[P: Table 5 and Project Management. Q: §6.2. S: §§4.2.5–4.2.6.]

The SRS requires a vertical slice before breadth: a small board runs from brief to BOM using the real interface contract, engine, backend and live distributor data, and is demonstrated at M4. Subsequent work widens and refines that slice. M4 is in Week 11; this plan does not create an earlier formal integration milestone. The Quality Plan additionally requires the analysis views to be checked at M4 using seeded completed runs, so an empty analytics surface is identified before final validation.

Development is trunk-based with short-lived branches, non-author review and mandatory CI gates. Supplier/model calls remain outside the pure engine. Normalised physical specifications may be reused through the documented cache, while stock, lifecycle and datasheet links are read live for each new run. Stored snapshots and recorded fixtures support repeatable replay rather than pretending to be live commercial data.

[P: Project Management. Q: §§5.2, 5.2.1 DS-5 and 6.2 M4. S: §§4.2.5, 8.3 and 8.5.]

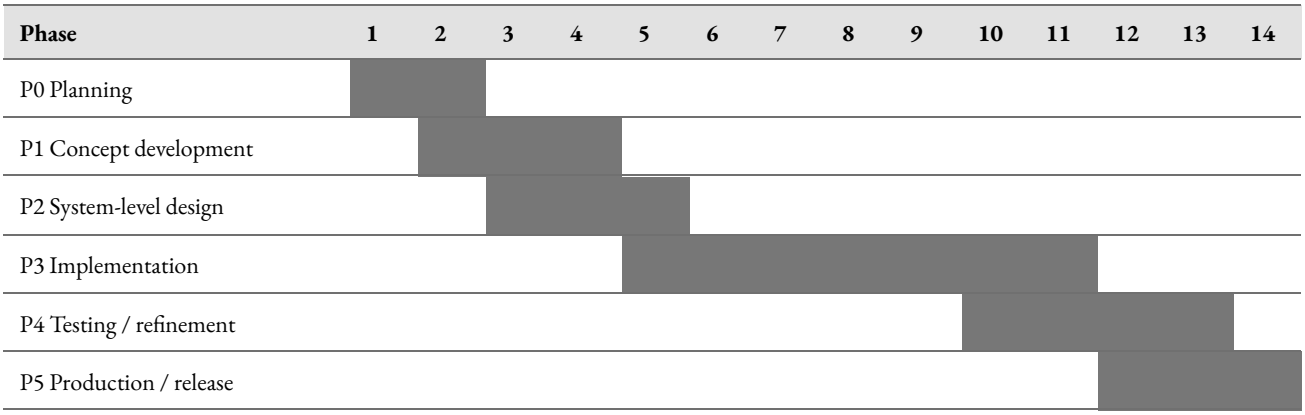
4 Schedule

4.1 Activity Dependencies and Schedule

The baseline period is Thursday 20 August to Wednesday 25 November 2026. Weeks run Thursday–Wednesday, as implied by the proposal's start/end dates and Wednesday milestone dates. Calendar ranges below are derived from that week structure. They are phase windows, not timesheets, exact subtask estimates or evidence of completion.

Week	Calendar range	Week	Calendar range
1	20 Aug–26 Aug 2026	8	08 Oct–14 Oct 2026
2	27 Aug–02 Sep 2026	9	15 Oct–21 Oct 2026
3	03 Sep–09 Sep 2026	10	22 Oct–28 Oct 2026
4	10 Sep–16 Sep 2026	11	29 Oct–04 Nov 2026
5	17 Sep–23 Sep 2026	12	05 Nov–11 Nov 2026
6	24 Sep–30 Sep 2026	13	12 Nov–18 Nov 2026
7	01 Oct–07 Oct 2026	14	19 Nov–25 Nov 2026

Figure 1. Phase-level Gantt schedule (filled cells indicate the proposal's scheduled phase window)



[P: Table 5 and Figure 4. Calendar week boundaries are calculated from the stated 20 August start. Only the documented phase weeks are reproduced, not task-level estimates.]

The phase windows overlap. P3 includes work in Week 5, while the formal contract milestone is at that week's end; preparation is not interpreted as prohibited before M2. Contract-dependent integration must use the frozen baseline. The sources provide no detailed working-day calendar or approved duration for each smaller task.

Milestones and phase gates

ID	Milestone / required result	Target date	Week	Depends on
M1	Requirements baseline approved	2 Sep 2026	2	—
M2	Interface contract frozen	23 Sep 2026	5	M1
M3	Constraint engine complete and verified	14 Oct 2026	8	M2
M4	End-to-end vertical slice live on real data	4 Nov 2026	11	M3
M5	Validation complete; specification evidence reviewed	18 Nov 2026	13	M4
M6	Final delivery, demonstration and handover	25 Nov 2026	14	M5

[P: Table 6. Q: §6.2. All dates above are targets; sponsor acceptance evidence was not supplied.]

The D3 System Design Document is separately due on 30 September, after M2. D9 deployment is due on 11 November. The Quality Plan requires M4 to include live-data integration, sourceability, repair, streaming, cache timing, explicit client fallback, CSV export and analytics against a seeded corpus. M5 reviews D6–D8; M6 closes the final evidence and handover.

Availability gate limitation: deployment on 11 November provides only one week before M5, although DS-12 requires a two-week observation. The Quality Plan gives two alternatives: begin observation by 4 November to close at M5, or record DS-12 as pending until 25 November and seek sponsor agreement on gate timing. Neither alternative is asserted as already approved here; the two-week requirement is unchanged.

[P: Table 8. Q: §§6.2 and 14.]

4.2 Work Breakdown Structure

The following work breakdown translates the source phases, modules and named responsibilities into eight work packages. X01–X08 and their sub-identifiers are labels created for this plan. They do not add features or replace the proposal's P0–P5 phases, M1–M6 milestones or D1–D12 deliverables.

Work package	Documented work decomposed under it
X01 Project Planning and Requirements	X01.1 Elicitation and interviews X01.2 Freedom-to-operate review and requirements baseline X01.3 Schedule, budget, risks, change control and sponsor coordination
X02 Concept Development	X02.1 Distributor field-coverage evaluation X02.2 Model selection / benchmark X02.3 User journey and executable acceptance scenario
X03 System Architecture and Interface Contract	X03.1 Board graph and part specification with provenance X03.2 API/event schema, sequence and recovery contract X03.3 Module design, rule formulas and resolution policy documentation
X04 Constraint Engine and Resolution Policy	X04.1 Ten pure rules and missing-data handling X04.2 Legal repair options, ranking and bounded repair policy X04.3 Independent engine verification and documentation
X05 Backend, Data and Orchestration	X05.1 Sourcing adapters, normalisation and physical-property cache X05.2 Planning, repair/escalation and checkpointed orchestration X05.3 APIs/SSE, accounts, persistence, BOM and datasheet tools X05.4 Recorded findings, memory and read-only analytics queries
X06 Frontend Application	X06.1 Account/project workflows and walkthrough X06.2 Board graph, trace, conflict and BOM views X06.3 SSE sequencing, escalation, recovery and explicit failure states X06.4 Upload/export, saved boards, memory and analytics X06.5 Accessibility, usability and handover support
X07 Verification, Validation and Integration	X07.1 Rule/property, contract, integration and adversarial testing X07.2 Regression, injection and seeded analytics corpora X07.3 Live vertical slice, browser acceptance, usability and performance X07.4 Bench measurements and evidence consolidation
X08 Deployment, Release and Handover	X08.1 Repository/CI, fixtures, environments and secrets X08.2 Hosting, schema-change procedure, monitoring and local fallback X08.3 Release archives, runbook, user guide, final report and demonstration

[P: Tables 5, 7 and 8; Figure 4. Q: §§3–4, 6–7, 9–14. S: §§4.2.5–4.2.6, 7–12. Bench resource gaps remain explicit in Sections 5 and 10.]

4.3 Work Packages

ID	Work package	Coordination / supporting roles	Scheduled window
X01	Project Planning and Requirements	Mohika; all members contribute; Sparsh/QA review.	P0 Weeks 1–2; management continues through Week 14.
X02	Concept Development	Sparsh coordinates technical decisions; Armaan, Lai Yi and Faye contribute within their roles.	P1 Weeks 2–4.
X03	System Architecture and Interface Contract	Sparsh; Armaan/Lai Yi and QA review interfaces and evidence.	P2 Weeks 3–5; D3 documentation due 30 Sep.
X04	Constraint Engine and Resolution Policy	Sparsh; Samarth verifies.	P3 through M3, 14 Oct; refinement continues as needed.
X05	Backend, Data and Orchestration	Armaan; Mohika component data and caching (X05.1); Sparsh technical review, Jerral infrastructure, QA tests.	P3 Weeks 5–11; P4 refinement.
X06	Frontend Application	Lai Yi; Armaan contract/API dependencies, Faye validation.	P1 storyboard; P3 build; P4 refinement.
X07	Verification, Validation and Integration	Samarth verification; Faye validation; Armaan/Lai Yi demonstrate M4; developers correct defects.	Throughout development; P4 Weeks 10–13 is the main final-testing window.
X08	Deployment, Release and Handover	Jerral release; Mohika handover; Faye/Lai Yi training and team artefacts.	Early repository/CI work; P5 Weeks 12–14 for release.

Where a package combines several documented activities, the coordination assignment follows the proposal's technical authority and role responsibilities. The documents do not allocate a separate person-hour budget to each package. Section 5 therefore preserves the role-level estimates without treating them as independent package estimates.

[P: Tables 5 and 7. Q: §§3.3, 6.2 and 13. X02 coordination is a role-based inference from Sparsh's technical direction responsibility, not a separately recorded assignment.]

4.4 Activity Dependencies

Dependencies below distinguish the start of work from the evidence needed to complete integration. The frozen contract enables parallel work; it does not mean that the frontend must wait for the whole backend or that tests start only after coding finishes.

Package	Required inputs / dependency	Completion relationship
X01	Sponsor request and documented customer investigation.	M1 provides the requirements basis for M2.
X02	Customer needs and rule-field requirements; overlaps X01/P0.	Source-selection memo and acceptance scenario support detailed design.
X03	Requirements baseline; X02 source and user-journey findings.	M2 freezes the shared contract; D3 follows on 30 Sep.
X04	Board/part types, rule definitions and shared contract from X03.	Verified engine at M3 is the proposal's predecessor for M4.
X05	Shared contract; source adapters/data; engine interface and verdicts; checkpoint persistence.	Engine, backend, frontend and live data converge at M4.
X06	Shared board/part/verdict/API/event contract; backend data and streams for integration.	M4 demonstrates live client workflows, export and seeded analytics.
X07	Requirements/rule definitions for test design; components and fixtures as they become available.	Runs alongside X04–X06; M5 consolidates acceptance evidence.
X08	Source, configuration, schema-change instructions and release evidence.	CI/setup starts early; release readiness and final handover close at M6.

Figure 2. Activity network at milestone level

Requirements	Shared contract	Parallel build / verification	Integration	Acceptance	Handover
X01 / X02 M1 →	X03 M2 →	X04 engine → M3 X05 backend X06 frontend X07 tests alongside →	X05 + X06 + X07 M4 →	X07 M5 →	X08 + team M6

X08 setup and P5 release preparation overlap development/testing; the final column is handover, not the first deployment activity. The M1→M2→M3→M4→M5→M6 chain is the proposal's milestone dependency chain. The critical path, float and remaining task estimates are given in Section 4.6.

[P: Tables 5–6. S: §4.2.5. Q: §§6–7 and 9–10. Package dependency grouping is inferred from these stated interfaces and gates.]

4.5 Work Package Details

Each package uses the sample's fields: assigned personnel, effort, start, purpose, inputs, activities and outputs. An "earliest phase window" is a source-derived planning window, not an invented actual start date. Outputs reference existing deliverables; no additional contractual D-number is introduced.

X01 — Project Planning and Requirements

Field	Work package description
Project / package	FUZE / X01
Assigned to	Mohika leads project management; all seven members contribute to requirements and review within their documented roles.
Effort	No package-specific estimate supplied. Mohika's total project allocation is 112 hours; this also covers management outside the initial requirements work.
Start / duration	P0: Weeks 1–2, 20 Aug–2 Sep 2026 (two-week phase window). Project management continues across the fourteen-week period.
Purpose	Establish the requirements basis and maintain the schedule, resource, risk, communication and approval controls.
Inputs	Sponsor request, reported customer interviews, proposal scope/DS criteria and existing use-case/SRS material.
Activities	Elicit and review requirements; document interview findings; complete the scheduled freedom-to-operate review; establish tooling/repository arrangements; prepare the requirements baseline; coordinate milestones, status, risk, budget and changes.
Outputs	D1 requirements specification, interview findings and freedom-to-operate memo at M1; Project Plan and continuing schedule/risk/decision records. No separate Project Plan submission date is provided in the four main sources.
Exit / evidence	Requirements baseline demonstrated for M1 and written sponsor decision recorded. Actual sign-off and the completed freedom-to-operate memo were not supplied.

[P: Patent Search; Tables 5–8 and 10. Q: §§3.1.1 and 6.2. S: §§4.1–4.2.1.]

X02 — Concept Development

Field	Work package description
Project / package	FUZE / X02
Assigned to	Sparsh coordinates technical direction (role-based inference). Armaan contributes data-source evaluation; Lai Yi contributes the user journey; Faye contributes acceptance scenarios.
Effort	No package-specific person-hours or person-days supplied. This work draws on the role allocations in Section 5.2.1.
Start / duration	P1: Weeks 2–4, 27 Aug–16 Sep 2026 (three-week phase window). No more detailed task starts are supplied.
Purpose	Establish viable component sources, model choice and an inspectable primary user journey before full implementation.
Inputs	Customer needs, the ten rules' actual field requirements, proposal technology considerations and use-case workflows.
Activities	Evaluate distributor coverage against required parameters; select and benchmark the model provider and use a region-compatible endpoint; storyboard the primary journey; write an executable acceptance scenario that initially fails. The documented choices are JLCPCB, Mouser/manufacturer datasheets and the GLM-5.2 model.
Outputs	Accepted source-selection memo and one written, failing executable acceptance scenario; concept findings feeding D2/D3 and the subsequent build. These are P1 exit artefacts, not new numbered deliverables.
Exit / evidence	P1 closes on the exit conditions in proposal Table 5. No accepted memo or scenario execution result is supplied as evidence here.

[P: Table 5, Technology Consideration and Figure 4; Table 7 for role inference. Q: §11. S: §§5.2 and 11.6–11.7.]

X03 — System Architecture and Interface Contract

Field	Work package description
Project / package	FUZE / X03
Assigned to	Sparsh leads. Armaan and Lai Yi contribute backend/client interface needs; Samarth and Faye review testability; Jerral contributes configuration and release needs.
Effort	No package-specific estimate supplied. Sparsh's 196 project hours also cover engine and broader technical review duties.
Start / duration	P2: Weeks 3–5, 3–23 Sep 2026 (three-week phase window). M2 is 23 Sep; D3 is due 30 Sep.
Purpose	Define the shared contract and architecture needed for parallel, traceable implementation.
Inputs	Requirements and use cases; source-field needs; board, slot, part, rail, verdict and run concepts.
Activities	Define typed part specifications with optional fields and provenance; board graph; API surface; event types, sequence ordering and recovery; module boundaries; ten rule formulas, stated assumptions, resolution policy and read-only analytics path. Carry unresolved fields/policies forward explicitly.
Outputs	D2 machine-readable JSON Schema, type definitions, interface surface and ordering guarantees; D3 architecture, module/data-flow design, rule formulas and assumptions, resolution policy and rationale.
Exit / evidence	M2 contract review/freeze, followed by D3 delivery. The frozen event contract is in the repository (backend/fuze/api/events.py and frontend/src/app/lib/types.ts); its approval record is still to be filed.

[P: Tables 5–8 and architecture description. Q: §§4.2 and 6.2. S: §§4.2.2, 8.1–8.2 and 9.6.]

X04 — Constraint Engine and Resolution Policy

Field	Work package description
Project / package	FUZE / X04
Assigned to	Sparsh leads implementation/policy. Samarth independently verifies arithmetic, rule outcomes, determinism, missing-data behaviour and model fencing.
Effort	No separate engine-package estimate supplied. Sparsh's and Samarth's role totals cannot both be charged entirely to this package because they have other duties.
Start / duration	Implementation lies in P3 (starting Week 5); M3 completion/verification is 14 Oct, Week 8. This bounds a Weeks 5–8 window, not a precise task duration.
Purpose	Deliver the independently runnable deterministic engine and the legal, bounded resolution policy.
Inputs	D2 board/part/verdict types, provenance fields, rule definitions, stated assumptions and resolution policy.
Activities	Implement R1–R10 without network/model/clock access; preserve all verdict evidence; return unchecked for missing inputs; revalidate the whole board after placements/repairs; compute and rank legal actions; enforce model membership checks, timeout/illegal-choice fallback, three repairs per slot and the SRS added-slot cap.
Outputs	D4 standalone engine source with test suite and documented programmatic interface; relevant D6–D8 test/coverage/evidence contributions.
Exit / evidence	M3 independently runnable engine; all ten rules verified; DS-1–DS-4, DS-7 and DS-10 evidence with coverage checks. Ambient temperature, converter efficiency, build quantity and relaxation-policy questions remain subject to the SRS open issues.

[P: Tables 1–2 and 6–8. Q: §§5.2.1 and 6.2 M3. S: §§7.4, 8.3–8.4, 8.7–8.9; OI-16–OI-18 and OI-21.]

X05 — Backend, Data and Orchestration

Field	Work package description
Project / package	FUZE / X05
Assigned to	Armaan leads. Mohika builds the component data adapters, normalisation and cache (X05.1); Sparsh reviews technical/contract changes; Jerral supports environments, persistence deployment and fixtures; QA verifies and validates interfaces.
Effort	Armaan's total allocation is 182 hours across all backend responsibilities. A separate package/task split is not provided.
Start / duration	P3: Weeks 5–11, 17 Sep–4 Nov phase window; contract-dependent integration uses M2. P4 refinement continues through M5. Remaining task estimates are in Section 4.6.
Purpose	Connect requirement planning, real component data, deterministic validation, bounded repair and persistent streamed runs.
Inputs	D2/D3, engine interface, distributor/model endpoints, manufacturer datasheets, persisted project/run structures and test fixtures.
Activities	Implement JLCPCB/Mouser adapters through the MCP tool layer and provenance-preserving normalisation; physical-property cache keyed by MPN and prompt fingerprint; live stock/lifecycle/datasheet retrieval per run; planner and LangGraph orchestration; legal repair/escalation; checkpoint persistence; ordered SSE; authentication/project isolation; BOM upload/export and datasheet extraction; memory and read-only analytics queries.
Outputs	Backend portion of D5; persisted board/run/event/finding data; integrated APIs/SSE; source/timing/security/analytics evidence for D6–D8.
Exit / evidence	M4 live-data vertical slice and DS-5, DS-6, DS-9 evidence; DS-13 and DS-16 contributions. POST /resume handles interrupt answers; POST /threads/{id}/continue handles abandoned/error runs. These paths must remain distinct.

[P: Tables 4, 7–8. Q: §§3.3.2, 5.2.1 DS-5/DS-16 and 6.2 M4. S: §§7.5–7.12, 8.1–8.6, 8.9–8.12 and 11.]

X06 — Frontend Application

Field	Work package description
Project / package	FUZE / X06
Assigned to	Lai Yi leads. Armaan supplies APIs/data/streams; Sparsh owns contract/engine decisions; Faye validates browser and usability outcomes. Jerral helps with stream error and stall handling and retains release/infrastructure duties.
Effort	Lai Yi's total project allocation is 182 hours, including implementation, refinements and supporting evidence. No task-by-task allocation is supplied.
Start / duration	P1 storyboard precedes the build. Main implementation is in P3, Weeks 5–11; the live vertical slice is M4 on 4 Nov. P4 refinement targets M5; D11 support continues to M6.
Purpose	Provide an inspectable, accessible client for the documented user workflows without misrepresenting missing data, failed operations or incomplete runs.
Inputs	Use cases/SRS, D2 types/events, backend APIs and streams, persisted board/verdict records and a seeded analytics corpus.
Activities	Build account/session and project screens, walkthrough, design workspace, component graph, validation/conflict views, BOM table, escalation answers and stopped-run recovery; upload/export and datasheet controls; read-only saved-board/memory/analytics views; ordered SSE rendering and duplicate suppression; explicit blank/stall/error states with last sequence identifier and retry/resume control. Refine accessibility/usability and verify exported files.
Outputs	React 19/TypeScript frontend in D5; client evidence for DS-6, DS-11, DS-15 and DS-16; D11 user-guide/training contributions.
Exit / evidence	M4 live integration, honest error/empty states, export and seeded analytics; M5 usability and browser acceptance. Analytics rates show numerator, denominator and unchecked count; withheld rates show counts/reason, not a false zero. Export-state policy is a source conflict recorded in Section 10.

[P: Table 7. U: UC-1–UC-16. Q: §§3.3.2, 5.2.1 DS-6/DS-11/DS-15/DS-16, 6.2 and 13. S: §§7–9.]

X07 — Verification, Validation and Integration

Field	Work package description
Project / package	FUZE / X07
Assigned to	Samarth leads verification; Faye leads validation and the acceptance pack. Armaan/Lai Yi demonstrate M4. Developers fix defects; Jerral records fixtures/environment/release evidence. A bench-test activity owner is not specifically named.
Effort	Samarth and Faye each have 154 total project hours. A suite-, integration- or bench-specific effort split is not provided.
Start / duration	Test design and verification run throughout development. P4 is Weeks 10–13, 22 Oct–18 Nov. M4 integration is 4 Nov; M5 reviews D6–D8; any incomplete DS-12 window remains open to the agreed gate.
Purpose	Demonstrate requirements through repeatable tests and independently inspectable acceptance evidence.
Inputs	DS-1–DS-16, UC-1–UC-16, D2/D3, engine/application builds, recorded distributor fixtures and known expected outcomes.
Activities	Run rule/property, contract, integration, regression, adversarial, browser, usability and performance checks; maintain 40-board, 10-unsatisfiable-board, 25-injection-brief and at-least-20-completed-run analytics test data; conduct ten first-time-user sessions; compare bench measurements with thermal/current predictions; preserve failures, retests and regression tests.
Outputs	D6 suite/coverage report, D7 rerunnable procedures, D8 validation evidence; milestone readiness and open-defect reports.
Exit / evidence	Every required criterion evidenced for release; mandatory regression/CI checks pass; no unresolved false-pass, silent-blank-client or critical security defect. Bench equipment, measurement procedures/tolerances, procurement and detailed ownership are not supplied and require confirmation.

[P: Figure 4 and Tables 1, 7–8. Q: §§5–8 and 14. S: §§4.2.6 and 10.4.]

X08 — Deployment, Release and Handover

Field	Work package description
Project / package	FUZE / X08
Assigned to	Jerral leads release engineering. Mohika coordinates sponsor acceptance/handover; Faye and Lai Yi support user training; all members supply their final artefacts.
Effort	Jerral's total allocation is 126 hours. Handover also uses existing team allocations; no additional package budget is specified.
Start / duration	Repository/tooling begins in P0. P5 release window: Weeks 12–14, 5–25 Nov. D9 deployment is due 11 Nov; final handover is 25 Nov.
Purpose	Provide a reproducible hosted application, release evidence and a documented path for operation, local fallback and handover.
Inputs	Application source, dependency/configuration records, environment/secret settings, schema-change procedure, test results and sponsor-nominated artefact locations.
Activities	Operate CI, fixture recording and release tags; deploy HTTPS application/static site and managed PostgreSQL; provision secrets outside version control; document the SRS idempotent startup schema-change and corrective reversal procedures; monitor uptime, error rate and run completion; preserve release archives/second copies and verify restore access; prepare runbook, guide, final report and demonstration.
Outputs	D9 deployed system; D5 source release; D10 runbook; D11 guide/training; D12 final report and 20-minute demonstration.
Exit / evidence	M6 handover and sponsor acceptance; complete availability observation and final evidence. Cold start ≤60 s, warm health response ≤500 ms and availability ≥99% over two weeks are targets, not measured results in these sources.

[P: Tables 7–10. Q: §§6.2, 9–13. S: §§6.4, 10.3 and 12.]

4.6 Critical Path and Prototype Status

4.6.1 Critical Path

The packages on the critical path are X01 → X03 → X04 → X05 → X07 → X08, through milestones M1 → M2 → M3 → M4 → M5 → M6. None of them has float: M6 is fixed at 25 November, so a slip in any of them moves the final delivery.

Package	Window	Duration	Float
X01 Planning and requirements	Weeks 1–2	2 weeks	0 (critical)
X02 Concept development	Weeks 2–4	3 weeks	1 week (needed by M2, end of Week 5)
X03 Interface contract	Weeks 3–5	3 weeks	0 (critical)
X04 Constraint engine	Weeks 5–8	4 weeks	0 (critical)
X05 Backend and orchestration	Weeks 5–11	7 weeks	0 (critical)
X06 Frontend	Weeks 5–11	7 weeks	0 planned; 4 weeks realised (prototype complete in Week 7)
X07 Verification and validation	Weeks 10–13	4 weeks	0 (critical)
X08 Release and handover	Weeks 12–14	3 weeks	0 (critical)

The tightest constraint is DS-12. It needs two weeks of uptime before M5, but D9 is due on 11 November, which leaves one week. The prototype is already deployed at https://continuity-ui.onrender.com, so the observation window will start by 21 October. That closes the DS-12 window two weeks before M5.

4.6.2 Prototype Status at Week 7

As of 7 October 2026, the following work end to end on the campus GitLab repository (branch continuity-import, with the analytics work under merge request into main): the ten-rule constraint engine; the planner, and JLCPCB and Mouser sourcing; the LangGraph repair loop, and FastAPI with the event stream; accounts and projects; the React client with the component graph, trace, conflict drawer and BOM table; and CSV export, standalone BOM validation, datasheet attachment, memory, the walkthrough and the UC-16 analytics dashboard.

Testing: 757 backend tests pass, with 91% backend coverage and 97% constraint-engine coverage, which meets DS-8. 17 Playwright browser tests pass.

Not yet in place: a GitLab CI pipeline enforcing the coverage gates; [Samarth: status of the 40-board regression and 25-brief injection corpora]; the ten usability sessions; bench measurements; and D6–D8.

4.6.3 Remaining Work Estimates

The development team estimated the remaining work from its experience building the prototype.

Task	Owner	Estimate	Due
M3 engine verification evidence	Samarth, Sparsh	4 days	14 Oct
GitLab CI pipeline with coverage gates	Jerral	3 days	21 Oct
Start DS-12 uptime observation	Jerral	1 day, then a 14-day window	21 Oct
Live vertical slice recorded for M4	Armaan, Lai Yi	3 days	4 Nov
Ten usability sessions	Faye, Lai Yi	10 days	18 Nov
D6–D8 test report, procedures and evidence pack	Samarth, Faye	7 days	18 Nov
D10 runbook and D11 user guide	Jerral, Lai Yi	5 days	25 Nov
D12 final report and demonstration	All; Mohika coordinates	7 days	25 Nov

5 Project Estimates

5.1 Code Size Estimation using Function Points

Code size is estimated with function points, following the sample plan. The team counted the functions from the sixteen use cases and the interfaces in the SRS, rated each one, applied the influence factors and converted the adjusted total into lines of code. The counts, ratings and scores below are the team's own assessment; they are not taken from the four source documents.

5.1.1 Unadjusted Function Points

Each function is classed as an input, output, inquiry, logical file or interface and rated Low, Average or High with the standard IFPUG matrices (data elements against files or record types referenced). The weights used are: inputs 3/4/6, outputs 4/5/7, inquiries 3/4/6, logical files 7/10/15 and interfaces 5/7/10.

Type	Function	Complexity
Input	Register account (UC-1)	Low
Input	Sign in (UC-1)	Low
Input	Sign out (UC-1)	Low
Input	Create project (UC-2)	Low
Input	Rename project (UC-4)	Low
Input	Delete project (UC-4)	Low
Input	Submit brief and start run (UC-6)	High
Input	Run demonstration design (UC-5)	Average
Input	Answer a question and resume (UC-9)	Average

Type	Function	Complexity
Input	Continue stopped run (UC-10)	Average
Input	Attach datasheet (UC-14)	High
Input	Upload or paste BOM for validation (UC-12)	High
Output	Live design run updates: plan, parts, results, repairs (UC-6 to UC-8)	High
Output	BOM CSV export (UC-13)	Average
Output	Run statistics (UC-16)	High
Output	Standalone BOM validation result (UC-12)	Average
Output	Datasheet values and updated result (UC-14)	Average
Inquiry	Check current session (UC-1)	Low
Inquiry	List projects (UC-3)	Low
Inquiry	List runs in a project (UC-3)	Low
Inquiry	View saved board (UC-11)	Average
Inquiry	Browse memory (UC-15)	Average
Logical file	Engineer account and session	Low
Logical file	Project	Low
Logical file	Run: brief, events, checkpoints, repair counts	High
Logical file	Board and bill of materials	Average
Logical file	Part data cache	Average
Logical file	Memory of parts and findings	Average
Interface	JLCPCB parts library (parameters, stock, price)	Average
Interface	Mouser (lifecycle status, datasheet links)	Low
Interface	Language model service (GLM-5.2)	Low
Interface	Manufacturer datasheets	Low

Submitting a brief, attaching a datasheet and uploading a BOM are rated High because each touches the project, the run and the part data together. The live run updates and run statistics are High outputs because they show derived values drawn from several files. The run record is a High logical file because it holds the brief, every event, the checkpoints and the repair counts.

Characteristic	Low	Average	High	Total
Inputs	6 × 3 = 18	3 × 4 = 12	3 × 6 = 18	48

Characteristic	Low	Average	High	Total
Outputs	0 × 4 = 0	3 × 5 = 15	2 × 7 = 14	29
Inquiries	3 × 3 = 9	2 × 4 = 8	0 × 6 = 0	17
Logical files	2 × 7 = 14	3 × 10 = 30	1 × 15 = 15	59
Interfaces	3 × 5 = 15	1 × 7 = 7	0 × 10 = 0	22
Unadjusted FP	**56**	**72**	**47**	**175**

Total unadjusted function points = 56 + 72 + 47 = 175

[U: UC-1–UC-16. S: §§7, 8.1–8.2 and 11.8. Classification, complexity ratings and counts are the team's assessment.]

5.1.2 Adjusted Function Points

Influence factor	Score	Basis
Data Communications	5	Browser talks to the server over HTTPS, plus a live event stream.
Distributed Functions	3	Engine, server, model and data sources are separate parts.
Performance	4	Results needed within 130 s, first update within 2 s.
Heavily Used Configuration	2	Runs on hosting that sleeps when idle.
Transaction Rate	2	Low traffic outside demonstrations and testing.
On-line Data Entry	5	Everything is entered online.
End-user Efficiency	4	Live updates, guided walkthrough and a 10-minute learning target.
On-line Update	4	Runs and the part cache are updated online and can be recovered.
Complex Processing	5	Ten rules, board-level checks and a repair loop.
Reusability	4	The engine can be used on its own.
Installation Ease	2	Needs a runbook, secrets and database setup.
Operational Ease	3	Health checks, monitoring and a local fallback.
Multiple Sites	1	One hosted site plus a local fallback.
Facilitate Change	3	Data sources can be swapped through the tool layer.
Total score	**47**	

Scores: 0 = no influence, 1 = insignificant, 2 = moderate, 3 = average, 4 = significant, 5 = strong.

Influence multiplier = total score × 0.01 + 0.65 = 47 × 0.01 + 0.65 = 1.12

Adjusted FP = unadjusted FP × influence multiplier = 175 × 1.12 = 196.00

[S: §§4.2, 6, 8 and 10–11 for the performance, streaming, deployment and reuse requirements behind each score. Scores are the team's assessment.]

5.1.3 Lines of Code

The engine and backend are written in Python and the frontend in TypeScript. The QSM Function Point Languages Table (Release 5.0) gives JavaScript an average of 47 lines of code per function point, and TypeScript is close enough to use the same figure. QSM has no figure for Python, so 47 is used for both. This will be compared with the real code size at M3.

Lines of code = 196.00 FP × 47 LOC/FP = 9,212 LOC

Comparison with the prototype, 7 October 2026: the source holds about 18,400 non-blank lines, 9,995 lines of Python in backend/fuze and 8,360 lines of TypeScript in frontend/src. This count includes comments, and excludes 8,644 lines of tests. That is about twice the estimate. The source is heavily commented, so part of the gap is not functional code. The estimate will be re-baselined at M4.

[P: Table 4. S: §§11.2–11.5. Conversion factor: QSM Function Point Languages Table, Release 5.0; the Python factor is a team assumption.]

5.2 Efforts, Duration and Team Size Estimation

The proposal estimates 1,106 person-hours over fourteen weeks with seven named team members. These are part-time role allocations, not seven full-time staff. Summing the stated weekly allocations gives 79 team hours per week; 79 × 14 = 1,106 hours. This arithmetic checks the proposal's total rather than creating a new estimate.

Measure	Source-derived value	Basis / limitation
Project duration	14 weeks; 20 Aug–25 Nov 2026	Proposal schedule. The inclusive calendar span is 98 days; working-day availability is not supplied.
Team size	7 members	Proposal Table 7.
Total planned effort	1,106 person-hours	Proposal Table 9.
Combined weekly allocation	79 person-hours/week	Calculated sum of 8 + 14 + 13 + 13 + 11 + 11 + 9.
Person-days / productivity rate	8 hours per day; 62 LOC per person-day	Used only for the function point estimate below. The rate is an industry benchmark (1997), not a project-specific measure.
Task-level duration / contingency effort	2% management and 3% contingency on the function point estimate	The role budget itself has no separate contingency. Remaining task estimates are in Section 4.6.

Function point estimate. The lines of code from Section 5.1.3 give a second, independent estimate, which is compared with the proposal's role budget:

Effort = 9,212 LOC / 62 LOC per person-day = 148.6 person-days (PD).

Duration = 3 × (effort)^{1}/3 = 3 × (148.6)^{1}/3 = 15.9 working days, or 3.18 weeks.

Team size = 148.6 PD / 15.9 days = 9.35 people, about 10 full-time staff.

Person-hours = 148.6 PD × 8 hours = 1,188.8 hours; adding 2% for project management (23.8) and 3% for contingency (35.7) gives 1,248.2 hours.

The formulas assume a full-time team; FUZE hassevenpart-timemembers.Thefunctionpoint estimate of 1,248.2 hours is 142.2 hours (12.9%) above the proposal's 1,106-hour budget. The budget in Section 5.3 is not changed; the gap is monitored at each milestone and re-estimated at M3 against the real code size.

[Calculated from Section 5.1.3. Production rate: industrial benchmarks (1997) of 31 LOC/PD (United States) and 62 LOC/PD (Canada); the higher figure is the team's choice because existing frameworks (FastAPI, LangGraph, React) reduce hand-written code. Budget: P Table 9.]

5.2.1 Distribution of Effort

Role / member	Hours/week	Weeks	Person-hours	Notional cost at $20 SGD/hour
Project Manager — Mohika	8	14	112	$2,240.00 SGD
Lead Developer — Sparsh	14	14	196	$3,920.00 SGD
Backend Developer — Armaan	13	14	182	$3,640.00 SGD
Frontend Developer — Lai Yi	13	14	182	$3,640.00 SGD
QA Verification — Samarth	11	14	154	$3,080.00 SGD
QA Validation — Faye	11	14	154	$3,080.00 SGD
Release Engineer — Jerral	9	14	126	$2,520.00 SGD
Total	79	14	1,106	$22,120.00 SGD

The same people support several work packages. Their role hours are therefore shown once, not duplicated under each package. The documents do not provide a percentage distribution across planning, design, coding, integration or documentation. No industry phase percentages are applied to the role budget; the management and contingency uplift is used only in the function point estimate in Section 5.2.

[P: Budget, Table 9. Q: §3.3.1. Weekly aggregate and inclusive calendar span are arithmetic derivations from these values.]

5.3 Cost Estimates

The table preserves the proposal's requested budget and notional loaded student-engineer rate of $20 per hour. Amounts are shown in Singapore dollars ($ SGD); the proposal writes them with "$" only. These are proposal estimates, not verified current vendor quotations, approved funding or incurred expenditure.

Item / supplier	Quantity / basis	Unit price	Estimated total
Personnel	1,106 hours	$20.00 SGD/hour	$22,120.00 SGD
Application hosting — Render	4 months	$10.00 SGD/month	$40.00 SGD
Managed PostgreSQL — Neon	4 months	$19.00 SGD/month	$76.00 SGD
Static site hosting — Render	4 months	$0.00 SGD	$0.00 SGD
Model inference credits — Z.ai (GLM-5.2)	1 allocation	$200.00 SGD	$200.00 SGD
Component parameters, stock and pricing — JLCPCB	1 allocation	$0.00 SGD	$0.00 SGD
Lifecycle and datasheet data — Mouser	1 allocation	$0.00 SGD	$0.00 SGD
CI minutes — GitLab CI (campus server)	1 allocation	$0.00 SGD	$0.00 SGD

Item / supplier	Quantity / basis	Unit price	Estimated total
Domain name — registrar unspecified	1 year	$15.00 SGD	$15.00 SGD
Cloud and data subtotal	As listed	—	$331.00 SGD
Non-personnel contingency	20% × $331.00 SGD	—	$66.20 SGD
Total requested estimate	$22,120 + $331 + $66.20 SGD	—	$22,517.20 SGD

[P: Table 9. Arithmetic is unchanged from the proposal.]

Resource and costing limits

The four sources do not provide costs, procurement lead times or confirmed access for the fabricated test board and bench instruments described in SRS Section 10.4 and the proposal Gantt chart. They also do not specify developer workstation purchase costs, actual supplier account terms or the registrar. These items are not silently entered as zero-cost resources.

SRS Section 11.10 describes components as open source or within free tiers except model credits; the proposal budget nevertheless includes paid hosting, database and domain allocations. This inconsistency is recorded in Section 10. The estimate above retains the proposal numbers without asserting licence clearance or changing its cost basis. The component-price USD default recorded in SRS Section 8.4 does not establish the currency of the overall project budget.

[S: §§8.4, 10.2–10.4 and 11.10. P: Figure 4 and Table 9.]

6 Product Checklist

The proposal identifies twelve numbered deliverables. Every item is to be delivered to the sponsor-nominated repository or shared drive by its due date, with receipt confirmed in writing by the Project Manager. The checklist below is a delivery baseline; no item is marked delivered or accepted without supporting evidence.

ID / deliverable	Required content	Due date / gate
D1 Requirements Specification	Functional/non-functional requirements traced to DS criteria, interview findings and freedom-to-operate memo.	2 Sep 2026 / M1
D2 Interface Contract	Frozen event schema, types, API surface and ordering guarantees in machine-readable JSON Schema plus prose.	23 Sep 2026 / M2
D3 System Design Document	Architecture, module/data flows, ten rules and formulas, stated assumptions, resolution policy and rationale; Q adds the analysis surface.	30 Sep 2026
D4 Constraint Engine Module	Standalone engine source, test suite, independent installation/execution and documented programmatic interface.	14 Oct 2026 / M3
D5 Complete Source Code	Backend, frontend and infrastructure configuration, layered commit history and detailed README.	25 Nov 2026 / M6
D6 Test Suite and Coverage Report	Unit, integration, adversarial and browser tests; DS-8 coverage; 40-board and 25-brief corpora; recorded distributor fixtures; Q adds seeded analytics tests/data.	18 Nov 2026 / M5
D7 Test Procedure Document	Setup, steps, expected result and rerun instructions for each acceptance test, executable independently by the sponsor.	18 Nov 2026 / M5
D8 Validation Evidence Pack	Measured results for all DS criteria, including usability, timing and analytics evidence; identify any pending DS-12 observation.	18 Nov 2026 / M5
D9 Deployed System	Live application at a public HTTPS address.	11 Nov 2026
D10 Deployment Runbook	Environment, secrets, schema changes, monitoring, corrective reversal/rollback and local-hosting fallback.	25 Nov 2026 / M6
D11 User Guide and Training	Brief writing, conflict/evidence interpretation, BOM upload, datasheet attachment and export.	25 Nov 2026 / M6
D12 Final Report and Demonstration	Approach, measured results for every specification, limitations and future work; 20-minute live demonstration and questions.	25 Nov 2026 / M6

[P: Deliverables, Table 8. Q: §4.2 and §§6.2–7. S: §12 for the schema-change procedure.]

P1 additionally requires the accepted data-source selection memo and a written failing acceptance scenario; these remain phase-exit artefacts rather than new D-numbered deliverables. A standalone Project Plan due date is not provided by the four main documents. The source-nominated repository/shared-drive addresses, actual submission records and acceptance signatures are also not supplied.

7 Best Practice Checklist

This checklist records practices required or proposed by the four sources. It is not a completed audit. "Evidence" identifies what the team should inspect before marking a practice satisfied; Section 4.6 records the state of the current prototype.

Practice	Required control / evidence to inspect
Requirements and document consistency	Trace requirements/use cases and DS-1–DS-16 to deliverables/tests; retain version, reviewer, change reason and decisions; carry unresolved issues explicitly.
Architecture and interface discipline	Typed data, documented units, provenance-bearing part fields, frozen JSON/event schema and ordering/recovery guarantees.
Deterministic validation	Rules have no network, model or clock access; repeated recorded inputs produce byte-identical verdict sets.
Honest missing-data / client behaviour	Unchecked never becomes pass; load failure differs from an empty result; incomplete runs and stalled/blank clients are visibly identified.
Model and repair controls	Validate choices against engine-computed legal actions; revalidate after changes; cap repairs and preserve counters across recovery.
Review before merge	Short-lived branches; at least one non-author engineering review; Sparsh reviews engine/contract changes.
Automated quality gates	Mandatory CI passes before merge; backend coverage ≥80% and engine coverage ≥95%; regression and outcome tests complement coverage.
Reproducible data / supplier control	Recorded fixtures for offline replay, normalisation prompt/version cache keys, explicit source/schema errors and supplier-change regression review.
Security and identity isolation	argon2id passwords, secure session cookies, per-engineer access checks, protected secrets and repository secret scans.
Inspectable analytics	Stored source records only; read-only computation; rates with counts and honest withholding; no persisted derived scores or cross-user analysis.
Configuration and release records	Pinned dependencies, versioned source/schemas/tests/configuration, release tags, schema-state records and rollback/local fallback instructions.
Evidence preservation	Milestone evidence archived beyond CI expiry, second release-evidence copy, restore access checked and one-year post-handover retention.
Training and independent checks	Record project-specific walkthroughs/exercises, non-author retests and sponsor-rerunnable acceptance procedures.

[P: Project Management and Table 1. Q: §§3.1.2, 5, 7–13. S: §§8.3–8.12 and 12.]

8 Risk Management

Mohika maintains the risk register. Samarth and Faye review it weekly with the team and at milestone gates; sponsor risk reviews follow the every-two-weeks communication schedule. Each register entry is to hold an owner, likelihood, impact, trigger, mitigation, contingency and review date. Correctness or milestone threats are escalated as soon as they are identified.

Likelihood and impact ratings, and the full register of seventeen risks, are kept in the Risk Management Plan, version 1.2. R01 to R09 below summarise its first nine risks.

[Q: §14. P: Table 10.]

Risk / owner	Exposure or trigger	Documented risk reduction / response
R01 Missing or unreliable part data Owner: Armaan	Required field absent, unreliable, conflicting or without defensible provenance; an unchecked case could otherwise appear validated.	Preserve field/value/source evidence; report missing values as unchecked; use recognised cited references or the documented datasheet path where available. Do not substitute numeric defaults.
R02 Normalisation drift Owner: Sparsh	Equivalent recorded inputs produce different normalised interpretations or verdict serialisations.	Freeze normalised inputs and serialisation; retain the 100-run determinism checks and provenance tests. Normalisation stays outside the engine.
R03 Illegal or looping repairs Owner: Samarth	Model action outside the legal set, timeout, repeated failed repair or a run that does not terminate.	Adversarial/legal-set tests; reject illegal or late choices and use the least-disruptive candidate; enforce three repairs per slot and escalation, with the SRS added-slot bound.
R04 Stream gaps / blank or stalled client Owners: Armaan and Lai Yi	Missing/duplicate events, no first render, blank canvas or stalled last event.	Persist/verify sequence and high-water marks; test interrupt resume and stopped-run continuation; show explicit error, last sequence ID and retry/resume control.
R05 Stale normalisation-cache interpretation Owner: Mohika	A changed normalisation prompt reuses an old physical interpretation.	Key the cache by MPN and prompt fingerprint so a changed prompt triggers parsing. Stock, lifecycle and datasheet links remain live on each run; fixtures are not the live runtime cache.

[Q: §14; supporting controls in §§5.2.1 and 7. S: §§8.3, 8.6–8.9.]

Risk register (continued)

Risk / owner	Exposure or trigger	Documented risk reduction / response
R06 Supplier rate limits / outages Owner: Armaan	Unavailable source, API/schema change, rate limit or source error during sourcing.	Use filtered queries and recorded fixtures for repeatable tests; verify adapters, units and provenance; fail explicitly on unexpected source changes and test timeouts/outages. Unfilled slots remain unresolved rather than falsely sourceable.
R07 Hosting sleep / outage Owner: Jerral	Cold start, failed health probe or hosted environment unavailable during demonstration.	Record health/availability probes and response times; preserve a documented local-hosting fallback. No recovery-time guarantee beyond the stated DS targets is provided.
R08 Insufficient usability participants Owner: Faye	Fewer than ten suitable first-time participants available before M5.	Recruit ten participants before M5 and record timed, unassisted task results under anonymous IDs. A separate recruitment contingency is not specified.
R09 Insufficient seeded analytics data Owner: Faye with Lai Yi	Fewer than twenty completed runs in the demonstration account four weeks before M4.	Prepare and commit a seeded corpus of at least twenty completed runs across several projects before M4; review analytics at M4 so thin/empty analysis is found before M5. Status, 7 Oct: done. tools/seed_analytics.py seeds 25 completed runs across five projects.

[Q: §14; supporting controls in §§5.2.1, 7 and 11. S: §§8.3, 8.6–8.9.]

The DS-12 observation-window conflict is an additional explicitly documented schedule risk: D9 on 11 November is too late for two full weeks before M5. The two source-prescribed responses are retained in Section 4.1 and require a recorded gate decision. Bench-resource gaps and unresolved requirement policies are maintained as open planning issues in Section 10; no unsupported probability or mitigation budget is attached to them.

A source or supplier change requires impact review, regression tests and updated evidence. Adding a third distributor is explicitly a scope change under the Quality Plan. Proposed acceptance/scope changes are reviewed technically by Sparsh and coordinated by Mohika for written sponsor approval; failed criteria must not be silently waived.

[Q: §§8, 11, 14 and 15.]

9 Quality Assurance

TheFUZESoftwareQualityAssurancePlan,Revision3.0,governsthedetailedqualityprogramme. This Project Plan summarises its responsibilities, tests and acceptance gates rather than replacing it. Samarth leads verification; Faye leads validation. The quality plan's own approval remains pending, and the listed criteria are targets and required evidence, not claims that tests have passed.

Every test record identifies the DS requirement, setup, input, expected outcome, execution steps, build and actual result. Evidence preserves failures, dependency errors, retests and relevant regression checks. D7 is written so the sponsor can rerun acceptance independently. Live integration tests complement recorded offline replay.

[Q: §§1.1, 3.1.2, 7 and 15. S: §4.2.6.]

Test suite	Scope and required test material	Main DS links
Unit / property tests	Each rule in isolation; passing, failing, boundary and missing-data cases; hand-calculated arithmetic and generated inputs.	DS-1–DS-4
Regression board corpus	Forty recorded boards and field-removal cases; repeated recorded data; ten unsatisfiable boards for repair termination.	DS-2, DS-4, DS-7
Integration tests	Sourcing, normalisation, cache, orchestration, authentication/isolation, persistence, SSE and analytics queries with known counts.	DS-5, DS-6, DS-9, DS-13, DS-16
Adversarial / injection corpus	At least twenty-five briefs; illegal/timed-out model choices, validation-override attempts and legal-set fencing.	DS-10, DS-14
End-to-end browser tests	The sixteen use cases through the real interface, including exports, interruption/recovery, malformed uploads and honest empty/error states.	DS-11, DS-15, DS-16
Usability sessions	Ten first-time participants, unassisted and timed; guided walkthrough, a conflict-bearing brief and CSV export in the task.	DS-15
Performance / availability	Live, recorded, warm/cold-cache timing; first rendered event; cold/warm health behaviour and two-week uptime observation.	DS-5, DS-6, DS-12

Coverage is separately enforced as DS-8. The SRS also calls for a fabricated test board and bench instruments to compare current/thermal predictions with measurements; the suite table does not supply the measurement procedure, tolerances, equipment list or specific bench owner. These remain open, not assumed complete.

[S: §4.2.6, Tables 2–4; §10.4. Q: §§5.2.1 and 7.]

Acceptance criteria — DS-1 to DS-8

ID	Acceptance criterion / evidence	Responsibility
DS-1 Rule coverage	At least ten rules, each with zero network/model/system-clock access. Evidence: rule registry, isolation tests and rule outcome tests.	Sparsh; Samarth verifies.
DS-2 Determinism	Byte-identical verdict sets across 100 consecutive runs with the same brief/cached part data. Freeze normalised inputs and serialisation; also exercise the brief-to-verdict path to expose normalisation drift.	Sparsh; Samarth verifies.
DS-3 Origin	100% of verdicts include deciding field name, verbatim value and a resolvable URL or named reference; zero null sources. Missing-field verdicts identify absence and source context without inventing a number.	Sparsh; Samarth verifies.
DS-4 No defaults	At least forty boards exercise missing required fields. Affected rules return unchecked and name the missing field; no substituted electrical/thermal/stock values.	Sparsh; Samarth verifies.
DS-5 Latency	For boards up to twelve slots, validated BOM within 130 s with live data and 35 s with recorded data. Report cold-cache, warm-cache and fixture runs separately. About 90 s warm-cache is an internal control, not a replacement gate.	Armaan; Faye validates.
DS-6 Streaming	First event within 2 s of submission; zero gaps/duplicates through interruption/resume/continuation. Compare rendered sequence with server event log; blank/stalled output shows an explicit failure, last sequence identifier and retry/resume control.	Armaan and Lai Yi; Faye validates.
DS-7 Bounded repair	No more than three repairs per slot before escalation; every run terminates on ten unsatisfiable boards. Test illegal/timed-out choices and least-disruptive fallback. SRS added-slot bound is also enforced.	Sparsh; Samarth verifies.
DS-8 Coverage	Backend executable-line coverage at least 80%; constraint-engine coverage at least 95%, measured with pytest-cov and enforced in CI. Record commit and exclusions.	Samarth; non-author review where Samarth authors the work.

[P: Table 1. Q: §5.2.1 and Appendix A. S: §§8.3–8.7. DS-4's known component-price currency default is separately recorded in SRS §8.4, not treated as an electrical/stock default.]

Acceptance criteria — DS-9 to DS-16

ID	Acceptance criterion / evidence	Responsibility
DS-9 Sourceability	Every BOM item maps to a real MPN with sufficient stock, or the shortfall is clearly flagged. Compare quantities and stock with captured source evidence.	Armaan; Faye validates.
DS-10 Model fencing	Every model-proposed repair is checked against engine-computed allowed options before application; zero unvalidated repairs accepted. Preserve rejection and revalidation evidence.	Sparsh; Samarth verifies.
DS-11 Export fidelity	Q requires the final post-repair BOM, not intermediate or failed output; include MPN, quantity, price, stock and datasheet URL. Check rows against final state and open in Excel/Google Sheets. U permits current/in-progress export: unresolved policy in Section 10.	Lai Yi; Faye validates.
DS-12 Availability	At least 99% over the two-week demonstration window; cold start ≤60 s, warm health response ≤500 ms. Preserve all probes/outages; missing samples count as failures. The observation gate timing remains unresolved.	Jerral; Faye validates.
DS-13 Security	argon2id passwords, secure session cookies, zero committed secrets and no cross-user resource access. Inspect configuration, negative access tests and secret-scan results; rotate exposed credentials before retest.	Armaan and Jerral; Samarth verifies.
DS-14 Injection resistance	At least twenty-five injection briefs produce zero false passes against independently defined expected rule outcomes.	Samarth; Faye validates.
DS-15 Usability	At least eight of ten first-time users independently run a design and identify the conflict cause within ten minutes. Record timing, outcome, assistance and observations; Q includes walkthrough and export in the task.	Lai Yi supplies build/script; Faye recruits and validates.
DS-16 Validation analytics	Across ≥20 seeded completed runs, reproduce every figure by direct queries. Display numerator, denominator and unchecked count; withhold a rule's rate when unchecked exceeds 20% of its denominator. No persisted derived score; exclude abandoned/error and other users' runs.	Prototype built by Sparsh in Week 7; Lai Yi with Armaan maintain it; Samarth verifies; Faye consolidates acceptance.

[P: Table 1. Q: §5.2.1, §§6.2–7 and Appendix A. U: UC-13, UC-16. S: §§8.8–8.12 and 9.4.]

The 20% analytics threshold is explicit in proposal DS-16, Q DS-16 and SRS Section 8.12, while U UC-16 and SRS OI-22 still carry confirmation language. The specified threshold is retained for the test baseline; the conflicting open-issue wording is not represented as closed. Rates must not improve merely because missing data is dropped.

Review gates, tools and corrective action

Review	Required quality focus
M1 — 2 Sep	Requirements, scope, resources, risks and testability of the initial DS baseline. Later DS-16 additions remain traceable.
M2 — 23 Sep	D2 schema/types, provenance and ordering; both QA managers review. D3 follows on 30 Sep.
M3 — 14 Oct	Standalone engine, all rules, arithmetic cross-checks, regression evidence and coverage.
M4 — 4 Nov	Live vertical slice, sourceability, repair, stream/recovery, cache timing, client fallback, CSV export and seeded analytics.
M5 — 18 Nov	D6–D8 and measured evidence for every DS criterion; explicitly identify incomplete DS-12 observation.
M6 — 25 Nov	Complete availability window/final evidence, D5 and D10–D12, demonstration, training and sponsor handover.

The documented testing/tool baseline is pytest, pytest-cov, Playwright, GitLab CI, JSON Schema checks, recorded distributor fixtures, HTTP/event logs and deployment probes. Secret scanning and dependency checks are CI controls. The client charting library must be pinned, but no particular library or exact test-tool version is named. Property-based cases and hand calculations complement example tests; no additional framework is selected in this plan.

Defects are recorded in the Quality Plan's proposed GitLab issues register with requirement/deliverable, severity, build/environment, steps, expected/actual result, evidence, owner and target milestone. The workflow is Open → Triaged → In progress → Ready for retest → Closed; failed retests reopen the issue. Critical, High, Medium and Low are defect severity categories, not preassigned risk probabilities.

QA triages weekly and immediately escalates false passes, security exposure, data loss and release blockers. Developers supply fixes and regression tests; a non-author retests before closure. Release requires all DS criteria evidenced, mandatory regressions passing and no unresolved false-pass, silent-blank-client or critical security defect. Acceptance changes require sponsor approval rather than an undocumented waiver.

[Q: §§6–9 and 15.]

10 Monitoring & Control

Monitoring compares the planned phases, milestones, role allocations and DS evidence with recorded delivery status. Progress at Week 7 is recorded in Section 4.6 and is updated at each milestone.

Control area	Documented method and responsibility
Schedule / deliverables	Mohika tracks milestones, dependencies and budget; confirms deliverable receipt; circulates review artefacts 48 hours early; records decisions/actions with owners and due dates. A phase remains open until its exit condition and sponsor sign-off are recorded.
Resources / costs	Use the 1,106-hour, role-based budget and proposal cost items as the estimate baseline. Actual hours and spending are recorded at each milestone, and remaining task estimates are in Section 4.6. Planned allocations are not consumption.
Quality / readiness	Samarth consolidates automated results on relevant CI runs; Faye consolidates acceptance evidence at M4–M6. Report failures/unexecuted checks, open defects by severity, ageing, recurrence and corrective-action status.
Risk / escalation	Mohika owns the register; both QA managers review weekly and at gates. Escalate correctness, security, data-loss and milestone threats immediately using the communication process.
Scope / acceptance changes	State reason, affected DS criteria/deliverables, test impact, effort, schedule and risk. Sparsh reviews technical effects; Mohika coordinates written sponsor approval for scope/acceptance changes. Update revisions, preserve the old baseline, notify the team and rerun affected checks.
Configuration / release	Jerral records dependency versions and schema state, tags milestone/release baselines and archives CI evidence. Source, schemas, tests, infrastructure and reviewable documents stay in version control; secrets stay in protected environments.
Supplier control	Armaan verifies adapter schemas, units, identities, provenance and rate-limit/error behaviour. Supplier changes require impact review, regression checks and updated evidence; a third distributor requires scope change control.
Records / access	Large evidence goes to the sponsor-nominated drive, linked by build/test ID. Keep a second release bundle and verify restore access. Samarth, Faye, Jerral and Mohika retain their evidence/decision custodianships; evidence is retained one year after final handover.

[P: Tables 7–10. Q: §§5.2.1, 6, 8, 10–12, 14–15.]

Project-specific training supports the controls: rule arithmetic/provenance; risk and change control; model fencing/injection tests; Git/CI/release; event ordering/recovery; reproducible tests; browser/usability sessions; and sponsor handover tasks. Attendance, exercises and follow-up needs are recorded. The documents specify these topics and leads, not completed training attendance.

[Q: §13.]

Source differences requiring a recorded decision or correction

These items are not silently harmonised. Where one source gives an explicit operational requirement and another retains an older/open statement, both positions are identified. This register does not assert new approvals or relax any DS threshold.

Topic	Source positions / remaining action
Availability and M5 timing	P Table 8 puts deployment on 11 Nov and P Table 6 puts all validation at M5 on 18 Nov. Q §§6.2/14 require the full two-week observation: start by 4 Nov or keep DS-12 pending to 25 Nov with sponsor agreement. No option is confirmed.
BOM export state and columns	U UC-13 permits the current BOM, including an in-progress run and honest unresolved/unchecked items. Q DS-11 requires final post-repair export only, excluding intermediate/failed runs. S §§7.5/9.4 also require honest findings/final fidelity, while OI-7 and the column-layout text leave row verdict/unchecked representation unsettled. Confirm allowed states and the common CSV schema.
Stopped recovery versus interrupt resume	S §4.2.4 and §8.1 explicitly distinguish UC-9 POST /resume from UC-10 POST /threads/{id}/continue for abandoned/error. U's contents and parts of UC-10 retain paused/interrupt wording. This plan follows the explicit SRS distinction; update the inconsistent use-case wording.
Analytics threshold and aggregation	P DS-16, Q DS-16 and S §8.12 specify withholding above 20%; U UC-16 and S OI-22 still call for confirmation. S §7.4 has four verdict outcomes, but the analytics descriptions emphasise pass/fail/unchecked. Detailed aggregation of warnings/accepted conflicts and conflict band definitions is not supplied; this is an inferred contract-detail gap, not a new metric.
Scope / maturity language	S OI-19–OI-20 request sponsor confirmation of memory, walkthrough and standalone BOM validation despite these appearing in the functional documents. S also describes some functions as implemented/shipping. The prototype implements these functions (Section 4.6); the sponsor's scope confirmation is still to be recorded.
Sponsor communication wording	P Table 10 and Q §8 use every-two-weeks Thursday sponsor contact, while P narrative mentions a weekly call. P requests a sponsor reply to technical queries within three working days; S §6.1 describes team support replies. Confirm the intended direction/cadence; no broader support SLA is assumed.
Bench validation resources	P Figure 4 and S §10.4 include a physical test board/bench comparison, but detailed owner, equipment access, cost, procurement lead time, procedure and tolerances are absent. Keep the activity visible and obtain the missing resource/test basis.
Budget / licensing / document status	P budgets paid services; S §11.10 broadly describes free/open-source use except model credits. Q approval remains pending; neither tests nor plan acceptance are evidenced. P's conclusion says fifteen specifications, while its Table 1 and Q/S include DS-16. Retain DS-1–DS-16 and do not assert procurement, licence or approval completion.

[P: Tables 1, 6, 8–10 and Conclusion. U: UC-10, UC-13 and UC-16. Q: §§5.2.1, 6.2, 8, 14–15. S: §§4.2.4, 5.9, 6.1, 7.4–7.5, 8.12, 9.4, 10.4 and 11.10.]

Open issues carried by the SRS — OI-1 to OI-14

The identifiers and owners below follow SRS Section 5.9. "Open in source" means no separate resolution record has been filed; it does not claim the issue remains unresolved in the current code. Some routes/input fields are described elsewhere in the same SRS, so contract confirmation is distinguished from inventing a route.

ID	Issue / confirmation required	Owner in SRS
OI-1	Whether registration automatically signs the engineer in or returns to sign-in.	Frontend and backend
OI-2	Paging, sorting or searching for large project/run lists.	Frontend and backend
OI-3	Whether project deletion removes its runs and retained findings/memory.	Backend
OI-4	Whether a server-side run continues after client disconnection.	Backend
OI-5	Whether an engineer can cancel an in-progress run.	Frontend and backend
OI-6	Whether graph and trace can be independently hidden.	Frontend
OI-7	Per-row verdict and unchecked-rule representation in exported CSV; reconcile with the export-state conflict above.	Frontend and backend
OI-8	Handling duplicate MPNs in standalone BOM validation.	Backend
OI-9	Whether standalone BOM validation findings persist and appear in memory.	Backend
OI-10	Whether the saved-board route always returns the latest persisted state.	Backend
OI-11	Response when a thread exists but has no saved board.	Backend
OI-12	Exact component identifier for datasheet attachment; S §7.10 requires MPN, package and PDF but the open issue is retained.	Backend
OI-13	Handling multiple candidate datasheets or component variants.	Backend
OI-14	Whether the datasheet route returns a file, reference, extracted values or combination.	Backend

[S: §5.9, OI-1–OI-14; §7.10. U: relevant Notes and Issues in UC-1, UC-3, UC-4, UC-6, UC-7 and UC-11–UC-14.]

Open issues carried by the SRS — OI-15 to OI-22

ID	Issue / status of source wording	Owner in SRS
OI-15	UC-10 naming and state mismatch. SRS explicitly follows "Recover a Stopped Run" for abandoned/error rather than interrupt resume; earlier labels/text require alignment.	Lead Developer
OI-16	Ambient temperature input for the thermal rule is undefined. Capture it in the brief or report unchecked; do not create a fixed numeric default.	Lead Developer
OI-17	Converter efficiency for the switching thermal calculation is usually not present in catalogue fields. Confirm its evidence/input path; missing input cannot be guessed.	Lead Developer
OI-18	Build quantity for stock comparison is not defined in the brief/input contract.	Lead Developer
OI-19	Confirm memory browsing as delivered scope with the sponsor and align the proposal. The source's shipping claim is not independent acceptance evidence.	Project Manager
OI-20	Confirm guided walkthrough and standalone BOM validation as delivered scope and align proposal traceability.	Project Manager
OI-21	When relax_requirement may be used and whether the engineer is informed that a requirement was weakened.	Lead Developer
OI-22	Confirm analytics rate-withholding policy. The >20% rule is explicit elsewhere, but this open-issue entry has no recorded closure. The prototype implements the 20% rule.	Lead Developer

[S: §5.9, OI-15–OI-22; §§8.4 and 8.12. P/Q: DS-16.]

Additional information not established by the four sources

Work-package or subtask effort and working-day durations; resource-levelled critical path/float; numeric risk probabilities or severity ratings; a separate Project Plan submission deadline; detailed board/part/event JSON schemas; standalone CSV structure, final export column order and concrete upload size limits; repository/deployment URLs; exact pinned dependency/charting-library versions; and actual expenditure and acceptance signatures are not provided. These remain identified gaps rather than invented plan content.

The Project Manager coordinates closure of the management/scope questions; technical, frontend/backend and QA responsibilities remain as named above. Any resolution that changes scope, acceptance criteria, schedule or resource assumptions follows the existing review/change process and is recorded in the plan's revision history. The plan is reviewed with the sponsor at each milestone.

[P: Tables 7–10. Q: §§6, 8, 9, 14–15. S: §§5.9, 7.9–7.10, 8.2 and 9.4. These details are added as they are agreed.]